

Lab 3: Component Modelling & Architectural Pattern Selection

PHARMACY EXPIRY & RE-ORDER ENGINE(15)

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WRITTEN JUSTIFICATION:-

Architecture Selection

Microservices Architecture

1. Architectural Choice

We chose **Microservices Architecture** for the Pharmacy Expiry & Re-order Dispatch Engine. The system is divided into independent components such as Inventory & Batch Manager, Expiry & FEFO Engine, Re-order Manager, and Supplier Dispatch Service.

2. Reasons for Selection

Reason 1 – Independent Services:

The expiry and FEFO processing can operate independently from inventory and re-order processing. This makes the system modular and easier to maintain.

Reason 2 – Scalability and Fault Isolation:

Individual services can be scaled independently according to workload. If one service experiences a failure, other services can continue operating without bringing down the complete system.

3. Security Advantage

Microservices can use **authenticated API communication and access control** between services. This ensures that only authorized users and services can access inventory and supplier-related operations.

4. Performance Benefit

Independent services allow **inventory checking, expiry monitoring, FEFO processing, and re-order operations** to run separately. Frequently used services can be scaled independently, improving system responsiveness.

Conclusion

The selected Microservices Architecture is suitable for the Pharmacy Expiry & Re-order Dispatch Engine because it supports modularity, independent scaling, fault isolation, and secure communication between system components.